

Counting (Homework)

① Gordon is buying a new custom guitar. He has the following choices to make. How many outcome variations are possible?

Color	# of Strings	Whammy Bar	# of Pickups
Sunburst	6-string	Yes	One Pickup
Powder Blue	12-string	NO	Two Pickups
Corvette Red			Three Pickups
Candy Green			

② Casey is choosing the type of grapes that he wants for a fruit basket that he will send to relatives. The store that sells the baskets allows gift givers to customize the grapes that they purchase. Casey can choose red, green, or black, seeds or seedless, and large, medium, or small grapes. How many outcome variations are there?

③ Joan is buying a new computer. She has the following choices to make. How many outcome variations are possible?

Processor	RAM Memory	Storage	Screen	Graphics Card
Intel	8 GB	500 GB	13"	Geforce 4090
AMD	16 GB	1 T	15"	Geforce 4080
	32 GB	2 T	17"	Geforce 4070
		3 T	19"	Geforce 4060
		4 T		Radeon 7900

④ The Vietnamese alphabet contains 29 letters. How many three letter codes are possible? **Note:** Repetition and permutations are allowed.

⑤ Six people are asked to choose their favorite month of the year. How many outcome variations are possible? **Note:** If Bill chooses December and Mark chooses April, this is different from a scenario where Bill chooses April and Mark chooses December. Also, they might all choose the same month. Consequently, repetition and permutations are allowed.

⑥ Nine people are asked to choose their favorite season: Fall, spring, summer, or winter. How many outcome variations are possible? Note: If Linda chooses the fall, and Dan chooses the spring, this is different from a scenario where Linda chooses the spring and Dan chooses the fall. Also, they could all choose the same season. Consequently, repetition and permutations are allowed.

⑦ The Vietnamese alphabet contains 29 letters. How many three letter codes are possible if no letter can be used more than once? Note: Permutations are allowed.

⑧ Six people are asked to choose their favorite month of the year. How many outcome variations are possible if we know that none of the people will choose the same month? Note: If Bill chooses December and Mark chooses April, this is different from a scenario where Bill chooses April and Mark chooses December. Consequently, permutations are allowed.

⑨ There are 32 professional football teams. Two people are asked to choose their favorite team. If we know that they will not choose the same team, how many outcome variations are possible? Note: Permutations are allowed.

⑩ 13 musicians are auditioning for a jazz band. Every musician is capable of playing any instrument, and there are 6 available positions: piano, double bass, trumpet, guitar, drums, and the saxophone. How many outcome variations are possible if no musician is allowed to play more than one instrument? Note: If Jerry plays the drums and Bob plays the piano, this is different from a scenario where Jerry plays the piano and Bob plays the drums. Consequently, permutations are allowed.

⑪ The president of Jolly Land will have to choose a Secretary of Defense, a Secretary of Commerce, a Secretary of Education, and a Secretary of Labor. There are 7 candidates that the president is considering for these positions. How many outcome variations are possible for the president's cabinet if we assume that every candidate is capable of doing any position? Note: If Larry becomes the Secretary of Defense, and Nathan becomes the Secretary of Education, this is different from a scenario where Larry becomes the Secretary of Education, and Nathan becomes the Secretary of Defense. Consequently, permutations are allowed. Repetition is not allowed because we assume that a person cannot hold more than one position.

⑫ How many ways can 8 coins be lined up?

⑬ Johnny has 12 different action figures. How many ways can he line them up on his windowsill?

⑭ There are 18 cats at the pound that need to be rescued. Abbey has a farm and she will pick 5 of the cats to catch mice. How many outcome variations are possible? Note: In this problem, permutations are not counted.

⑮ A candy store has 6 different colors of gumballs. If Matthew chooses 4 gumballs, each a different color, how many possible outcome variations are there? Note: In this problem, permutations are not counted.

⑯ Steven opened a pack of baseball cards that contained 16 different cards. He decided to give 5 of the them to his younger brother. How many possible outcome variations are there? Note: In this problem, permutations are not counted.

Answers

- ① 48 Outcome variations
- ② 18 Outcome variations
- ③ 600 Outcome variations
- ④ 24,389 codes
- ⑤ 2,985,984 Outcome variations
- ⑥ 262,144 outcome variations
- ⑦ 21,924 codes
- ⑧ 665,280 outcome variations
- ⑨ 992 outcome variations
- ⑩ 1,235,520 bands
- ⑪ 840 cabinets
- ⑫ 40,320 ways
- ⑬ 479,001,600 ways
- ⑭ 8,568 groups/combinations
- ⑮ 15 groups/combinations
- ⑯ 4,368 groups/combinations